

PAPER_726: UQFF Quantum Variable Documents: delta_def, f_TRZ, T_s, phi_j

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Abstract

Five quantum variable documents (including one duplicate f_{TRZ} entry) define: δ_{def} (metric deformation perturbation), f_{TRZ} (Bearden time-reversal zone factor enabling negentropic processes), T_s (string temperature), $\hat{\phi}_j$ (azimuthal direction vector in U_m string coupling). The f_{TRZ} factor underpins the Red Dwarf plasmoid stability described in Drawing 30 (TRZ Vortex).

1. Variable Registry

Variable	Value	Physical Role
δ_{def}	0.001	Metric deform: $g_{eff} = g_{\mu\nu}(1 + \delta_{def})$
f_{TRZ}	0.1	TRZ negentropic: $U_i \cdot (1 + f_{TRZ})$
T_s	10^4 K	String thermal: $E_{str} = k_B T_s \cdot n_{modes}$
$\hat{\phi}_j$	1.0 (unit)	Direction projection in U_m
f_{TRZ} (dup.)	0.1	Duplicate document reference (same value)

2. Bearden TRZ Framework

The time-reversal zone factor $f_{TRZ} = 0.1$ enables negentropic (overunity) processes in Bearden's vacuum engineering framework:

$$U_i = \lambda_I \cdot \frac{\rho_{SCm}}{\rho_{UA}} \cdot \omega_i \cdot \cos(\pi t_n) \cdot (1 + f_{TRZ})$$

3. Metric Deformation

$$g_{eff} = g_{\mu\nu} \cdot (1 + \delta_{def}) \approx g_{\mu\nu} \cdot 1.001$$

Small post-Newtonian correction beyond standard GR from vacuum deformation.

4. String Temperature

$$E_{str} = k_B T_s \cdot n_{modes} \quad T_s = 10^4 \text{ K}$$

References

- UQFF KB grok_share_ba508f76c8e.txt entry #76
- Session 176, v5.33